Swapnanil Bala

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swapportfolio.vercel.app | github.com/SwapnanilBala | linkedin.com/in/swapnanil-bala-854b722a7

EDUCATION

Northeastern University, Khoury College of Computer Sciences Sep 2025 – Dec 2027 (expected)

M.S., Data Science | GPA 3.4/4.0

Coursework: Foundations of AI, Natural Language Processing, Machine Learning & Pattern Recognition, Algorithms, Essentials of Data Science, Data Mining Techniques

Maulana Abul Kalam Azad University of Technology (MAKAUT) 2019 – 2023

B.Tech., Electronics and Communications Engineering

Certifications: Google Advanced Data Analytics Certificate; Meta Database Engineer Professional Certificate

PROJECTS

Robust Health | app.robusthealth.in | *Next.js, TypeScript, Supabase, Claude API, Pinecone* May 2026 – Present

- Designed the backend database and linked every part together myself, using AI-assisted coding.
- Built a fitness planner that gives each member a weekly training, nutrition and recovery plan, revised from their logs.
- Grounded the AI's advice in trusted sources (NIH fact sheets, ~870 exercises, 48 muscle pages) using retrieval (RAG); the right passage ranks first for 93% of 45 test questions and in the top 5 for 96%.
- Each AI plan adds up to 8 rules for that member, each citing its source, for about $0.001 a plan; a backup rule-based planner with no AI call works around injuries (knee, back, shoulder).
- Every change runs 1,312 automated tests, including a check that all 57 private API routes reject signed-out users.

Lagna Atelier | lagnaatelier.site | *TypeScript, Neon PostgreSQL, pgvector, Claude API* Jul 2026 – Present

- Designed the backend database and linked every part together myself, using AI-assisted coding.
- After testers found the readings stale, grounded them in 3 classical texts (RAG); feedback turned strongly positive.
- Switched the AI from Claude Opus to Haiku: a full PDF report went from ~3 minutes and $0.37 to about one minute and ~$0.006, with live progress shown while it writes.
- Added automatic checks on every reading in all 7 languages (script, banned words, tone, one retry on failure); rewording the prompt cut English tone failures from 9 of 27 notes to 2.

Orthopedic Clinic Web App | drbalaortho.com | *Vercel, Neon PostgreSQL* Apr 2026 – Jun 2026

- Designed the backend database and linked every part together myself, using AI-assisted coding.
- Built a fill-in-the-blanks prescription generator that a practicing orthopedic surgeon prints from every day.
- Built online booking and a patient portal in 3 languages: the doctor sets visit hours and approves patients, patients view and download past prescriptions, and every change is kept in an audit trail.

BRFSS Health Dashboard | brfssdashanalysis.vercel.app | *Python, Plotly Dash, Pandas* Nov 2025 – Dec 2025

- Turned a course R/Shiny app into a live dashboard of CDC health-survey data (2011-present, 3.2M rows): pick any question to see its trend, a state map and breakdowns by gender, age, race, education and income.
- Pre-computed every chart, turning the ~1 GB CDC file into 0.68 MB of chart data so the app runs as one lightweight function on Vercel; an automated check confirms the results match the original code.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, C++ (familiar)

ML / AI: Claude API, RAG, pgvector, Pinecone, prompt caching, PyTorch, scikit-learn, Pandas, NumPy

Web / Backend: Next.js, FastAPI, OAuth

Databases: PostgreSQL (Neon, Supabase), MySQL, database design

Tools: Git, CI/CD, Vercel, AI-assisted coding